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One-year double-blind study of the neurocognitive efficacy of olanzapine, risperidone, and haloperidol in schizophrenia☆

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Abstract

Neurocognitive deficits in schizophrenia can reach 1 to 2 standard deviations below healthy controls. The comparative effect of typical and atypical antipsychotic medications on neurocognition is controversial, and based primarily on studies with small samples and large doses of typical comparator medications. The present study assessed neurocognitive efficacy. It was hypothesized that olanzapine treatment would improve neurocognitive deficits to a greater degree than either risperidone or haloperidol treatment. This was a double-blind, randomized, controlled, parallel study with neurocognition assessed at baseline, and 8, 24, and 52 weeks. Per protocol, the haloperidol arm was discontinued. Four hundred and fourteen inpatients or outpatients with schizophrenia and schizoaffective disorder were treated with oral olanzapine ($n=159$), risperidone ($n=158$), or haloperidol ($n=97$). Individual domains (executive function, learning and memory, processing speed, attention/vigilance, verbal working memory, verbal fluency, motor function, and visuospatial ability) were transformed into composite scores and compared between treatment groups. At the 52-week endpoint, neurocognition significantly improved in each group ($p<0.01$ for olanzapine and risperidone, $p=0.04$ for haloperidol), with no significant differences between groups. Olanzapine- and risperidone-treated patients significantly ($p<0.05$) improved on domains of executive function, learning/memory, processing speed, attention/vigilance, verbal working memory, and motor functions. Additionally, risperidone-treated patients

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improved on domains of visuospatial memory. Haloperidol-treated patients improved only on domains of learning/memory. However, patients able to remain in treatment for the entire 52 weeks benefited more from olanzapine or risperidone treatment than haloperidol treatment.
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1. Introduction

Neurocognitive functioning is severely impaired in patients with schizophrenia, and performance on important components, such as verbal memory, executive functions, attention/vigilance, verbal fluency, information processing, and motor speed can reach 2 standard deviations (SD) below the mean of healthy control subjects (Braff et al., 1991; Goldberg et al., 1990; Harvey and Keefe, 1997; Heinrichs and Zakzanis, 1998; Javitt et al., 1995; Park et al., 1995; Saykin et al., 1991). These deficits appear to be a core element of the illness and not secondary to antipsychotic treatment or disease duration, as patients with first-episode psychosis have been shown to have severe deficits (Bilder et al., 1992; Bilder et al., 2000; Erlenmeyer-Kimling and Cornblatt, 1978; Harvey et al., 1998; Hoff et al., 1992; Saykin et al., 1994; Strauss and Klieser, 1990). Moreover, a growing body of evidence indicates that these deficits contribute to functional outcome impairment (Green, 1996; Green et al., 2000; Harvey et al., 1998; Velligan et al., 1997, 2000). Thus, pharmacotherapies that improve neurocognitive dysfunction will likely enhance the long-term outcome of patients with schizophrenia.

While atypical antipsychotic medications have generally been found to have a beneficial effect on neurocognitive deficits compared to typical medications (Bilder et al., 2002; Buchanan et al., 1994; Cuesta et al., 2001; Daniel, 1994; Gallhofer et al., 1999; Green et al., 1997; Hagger et al., 1993; Harvey et al., 2004; Hoff et al., 1992, 1996; Lee et al., 1994; Meltzer and McGurk, 1999; Purdon et al., 2000, 2001; Rossi et al., 1997; Sax et al., 1998; Stip and Lussier, 1996; Velligan and Bow-Thomas, 1999; Velligan et al., 2002), exceptions have been noted (Goldberg et al., 1993; Green et al., 2002). This previous work has suggested that atypical antipsychotics (olanzapine, quetiapine, risperidone, and ziprasidone) improve neurocognitive deficits in patients with schizophrenia, reduce indirect deleterious effects of extrapyramidal side effects (EPS) associated with typical antipsychotics, or allow a reduction of anticholinergic use and its associated cognitive impairment. However, many of these studies have had methodological weaknesses, including small sample sizes, nonblinded assessments, nonrandomization, inappropriate dosing, and limited trial duration. Recent studies with improved methodology, including large randomized samples with 2-year duration have suggested that lower doses of haloperidol may have a less deleterious impact on neurocognitive function (Green et al., 2002; Harvey et al., 2003; Keefe et al., 2004), that is equal to risperidone treatment in chronic schizophrenia patients (Green et al., 2002), although not as substantial as olanzapine (Keefe et al., 2004) or risperidone (Harvey et al., 2003) treatment in first-episode patients.

Little is known about the comparative effects of the various atypical antipsychotic medications on neurocognition. One early pilot study suggested that olanzapine treatment may have a greater neurocognitive benefit than risperidone or haloperidol treatment in early phase schizophrenia patients (Purdon et al., 2000). However, larger scale studies suggested that both olanzapine and risperidone were equally efficacious in improving cognition in schizophrenia (Harvey et al., 2003) and patients may benefit following a switch to ziprasidone from olanzapine, risperidone, or haloperidol (Harvey et al., 2004). Patients with treatment-resistant schizophrenia treated with high doses of olanzapine or risperidone have shown clear neurocognitive benefit compared to patients treated with high doses of clozapine or haloperidol (Bilder et al., 2002). Relative to these prior studies, this current study used a larger sample, fair comparisons with dosages aligned with current recommendations, non-treatment-resistant patients, and an extensive cognitive battery.

The present study assessed neurocognitive efficacy in 414 patients with schizophrenia and schizoaffective disorder, randomized to 52 weeks of double-blind treatment with olanzapine, risperidone, or haloperidol. It was hypothesized that olanzapine treatment would improve neurocognitive deficits after one year of treatment to a greater degree than both risperidone and haloperidol treatment.

2. Methods

2.1. Study setting and sponsorship

This project (F1D-MC-HGGN) was completed July 1999 through November 2000 at 39 sites in the United States and Canada with funding from Lilly Research Laboratories.

2.2. Patients

Patients enrolled in this study were 18 to 55 years of age, with a mean age of 39 years, who met the criteria for schizophrenia or schizoaffective disorder as defined in the DSM-IV, and at visit 1 had a minimum score of 4 on at least 2 positive items on the Positive and Negative Symptoms Scale (PANSS; scored 1–7; Kay et al., 1987) and a score of 18 or more (items scored 0–6) on the Brief Psychiatric Rating Scale (BPRS; Overall and Gorham, 1962; see Table 1); 40.6% of all patients had been previously admitted to the hospital in the past year due to psychiatric problems. Patients were to be English-speaking, have a level of understanding sufficient to agree to all tests and examinations, and had illness duration of at least 2 years from first hospitalization and/or diagnosis/treatment. Female patients of childbearing potential must have been using a medically accepted means of contraception.

Patients were excluded if they previously participated in the present study, participated in a clinical trial of another investigational drug within 1 month prior to study entry, or participated in a study within the past 3 months that included the neurocognitive battery. Other exclusion criteria were any significant neurological disorder, head injury with loss of consciousness, serious illness such that death was anticipated within 1 year or intensive care hospitalization was anticipated within 6 months, QTc interval greater than 450 ms, uncorrected hypo- or hyperthyroidism, current agranulocytosis, female patients who were either pregnant or nursing, allergic reaction to study medication, DSM-IV substance dependence (except caffeine and nicotine) within the past 2 months, treatment with depot antipsychotics, reversible monoamine oxidase inhibitor within 2 weeks, or clozapine or electroconvulsive therapy (ECT) within 1 month.

Concomitant drugs that were allowed chronically were antidepressants, except fluvoxamine and lithium. Acute usage of valproic acid, carbamazepine, antiemetics, and steroids was allowed. Benzodiazepines/hypnotics (<40 mg diazepam equivalent dose/day)

Table 1
Patient demographics and illness characteristics

Variable	Olanzapine $N=159$	Risperidone $N=158$	Haloperidol $N=97$	Overall $N=414$	p-values
Gender: male, n(%)	115 (72.3)	111 (70.3)	69 (71.1)	295 (71.3)	0.92[a]
Origin: n(%)					0.20[a]
Caucasian	95 (59.7)	101 (63.9)	51 (52.6)	247 (59.7)	
African	43 (27.0)	43 (27.2)	31 (32.0)	117 (28.3)	
Western Asian	2 (1.3)	0	0	2 (0.5)	
East/Southeast Asian	3 (1.9)	2 (1.3)	1 (1.0)	6 (1.4)	
Hispanic	13 (8.2)	6 (3.8)	9 (9.3)	28 (6.8)	
Other origin	3 (1.9)	6 (3.8)	5 (5.2)	14 (3.4)	
Mean age ± SD (years)	38.4 ± 7.90	39.5 ± 8.25	39.8 ± 8.32	39.1 ± 8.13	0.26[b]
Age of onset, mean ± SD (years)	23.0 ± 7.04	23.3 ± 7.19	22.6 ± 6.66	23.0 ± 7.00	0.98[b]
Diagnosis: n(%)					
Schizophrenia	105 (66)	117 (74)	68 (70)	290 (70)	0.30[a]
Schizoaffective	54 (34)	41 (26)	29 (30)	124 (30)	

[a] Frequencies analyzed using two-tailed Fisher's exact test.
[b] Means analyzed using Type III sum of squares from analysis of variance including treatment, investigator, and treatment-by-investigator interaction.

were allowed, but not concurrently, and were to be discontinued 48 h prior to neurocognitive testing. Benztropine mesylate or biperiden (up to 6 mg/day) were allowed throughout the study to control EPS, with dosing reduced as much as possible. Prophylactic use of anticholinergic medications was prohibited.

2.3. Study design and procedures

An informed consent document approved by the institutional review board or similar body was required to be understood and signed by all patients.

Patients (N=414) were double-blind randomized (1 : 1 : 1 ratio) to 52 weeks of treatment with olanzapine (5 to 20 mg/day), risperidone (2 to 10 mg/day), or haloperidol (2 to 19 mg/day). During the initial 8 weeks, a flexible dosing schedule was allowed; thereafter a fixed dose based on investigator's judgment was suggested. There was a 5 to 9 day screening period, in which initial screening, patient's history, psychiatric, and physical exams took place. Due to concerns regarding withholding medications, a washout period did not occur, thus patients were to continue with their normally prescribed medication until randomization at visit 2 (baseline). Patients could have either been treated with olanzapine or risperidone, but we could not systematically analyze this since medication dosages and time on treatment were highly variable. During the first 4 weeks, weekly visits were required, followed by 4 weeks of biweekly visits, after which patients were seen on a monthly basis. After approximately 52 weeks of enrollment the haloperidol arm was dropped due to recruitment difficulties. Patients currently enrolled were kept on current therapy and all future patients were randomized to either olanzapine or risperidone. The option of dropping the haloperidol arm due to slow enrollment was defined in the protocol. Reasons for discontinuation, use of benzodiazepines and anticholinergics, and compliance (noncompliance defined as intentionally missing more than 5 consecutive days of medication) were assessed at every visit.

2.4. Primary efficacy measures

The primary efficacy measure was change in a neurocognitive composite score derived from the neurocognitive battery. Patients were assessed at baseline, and 8, 24, and 52 weeks. The battery took approximately 90 min and was given in a consistent order. The following domains were assessed with the following tests, with outcome measures in parentheses. The measures were equally weighted within each domain unless indicated. Weighting of variables within each domain was based upon the authors' opinions about the relative importance of the measures collected, including the relationship of each measure to functional outcome. Executive Function, Trails B (time to complete, adjusted for Trails A time) and WCST 64-card version (Perseverative errors, Categories completed; Heaton, 1993); Learning and Memory, Rey AVLT with Crawford Alternative (Total words recalled [weighted * 2]; Words recalled after delay; Vakil and Blachstein, 1993) and Rey-Osterrieth Complex Figure (Immediate recall score [weighted * 2], Delayed recall score); Processing Speed, WAIS-R Digit Symbol (Total raw score [weighted * 2]), Trails A (time to complete); Attention/Vigilance, Continuous Performance Test (d-prime)(Keilp et al., 1997); Working Memory, WAIS-3 Letter-number Sequencing (Total correct items); Verbal Fluency, Controlled Oral Association Test (Total number of words generated) and Category Instances (Total number of words generated); Visuospatial Ability, Rey-Osterrieth Complex figure test (Copy score); and Motor Function, Grooved Pegboard (Total number of pegs). The composite score weighted each of the domains equally except executive functions and learning and memory, which were given twice the weight of the other domains due to their strong relationship to functional outcome and the number of tests used to assess them.

Three neurocognitive composite scores were developed based on baseline standards or empirically derived weights. Two of the composite scores were based on protocol-proposed domain weights and utilized all assessments. One was standardized by baseline data from the entire sample (sample composite), and the other was standardized from published norms (Spreen and Strauss, 1998; Crawford et al., 1989; Meyers and Meyers, 1995; Cornblatt and Keilp, 1994; Cornblatt, personal communication, 2000; WAIS-III, WMS-III Technical Manual, 1997; Wechsler, 1981; Heaton, 1993) with each controlling for age and gender. Standardized z-scores were computed by taking the patient's score, subtracting the corresponding mean of either the sample at baseline or the

normative mean and dividing by the SD for the corresponding measurement. Composite scores were then formulated by these averaged z-scores in patients for whom at least 75% of test scores were available.

2.5. Secondary efficacy measures

Psychopathology was measured by the PANSS, depression was assessed by the Montgomery–Asberg Depression Rating Scale (MADRS; Guy, 1976), and anxiety was assessed by the Hamilton Anxiety Scale (HAMA; Montgomery and Asberg, 1979). The proportion of patients who responded and patients who reached stability were compared across treatment groups. Response was defined as improvement in PANSS total score greater than or equal to 20% from baseline to any subsequent assessment. The definition of stability was a PANSS total score not increasing greater than 20% compared to baseline for at least one month. Among patients with baseline scores less than or equal to four and who achieved response or stability, the following relapse criteria were assessed using a score greater than four after any increase of greater than or equal to two (i.e., 3 to 5) on any PANSS positive items; or a score greater than or equal to four on an PANSS positive item and an increase of at least four on the PANSS positive subscale; or hospitalization.

2.6. Safety measures

A treatment-emergent adverse event (TEAE) was defined as an event that first occurred during the study or worsened after baseline. Frequencies of TEAEs were compared across treatment groups using a Fisher's exact test. Clinical laboratory testing was performed at baseline, and 8, 24, and 52 weeks. Vital signs were measured at every scheduled visit. Abnormal Involuntary Movement Scale (AIMS), Barnes Akathisia, and Simpson–Angus scales were assessed every week through week 4, every other week through week 8, and then every other month for the study duration.

2.7. Statistical analysis

Frequencies of baseline characteristics were analyzed by Fisher's exact test and means were compared using an analysis of variance (ANOVA). For the primary, protocol-specified, analysis, patients who completed baseline and one post-baseline neurocognitive assessment were included in a last-observation-carried-forward (LOCF) analysis of mean change from baseline to endpoint (8-, 24-, and 52-week). Secondary, observed case (OC) analysis, which included only those patients with non-missing composite scores assessed at the specific time period, was also conducted for all three endpoints. Patients were analyzed on an intent-to-treat basis. Analysis of covariance (ANCOVA) models for the neurocognitive analyses included the terms age, baseline, treatment, and investigator. Exploratory analyses without multiple comparisons were conducted on the individual assessments. Reasons for discontinuation, proportion of patients who were compliant, frequency of patients who used benzodiazepines or anticholinergics, AIMS, Barnes Akathisia, and Simpson–Angus scales were categorically analyzed between groups by Fisher's exact test. Mean changes in PANSS, MADRS, HAMA, weight, laboratory tests, AIMS, Barnes, and Simpson–Angus were assessed with Type III sum of squares ANOVA including terms for treatment and investigator. Differences in time to discontinuation were analyzed using the log-rank test from Kaplan–Meier's survival analysis.

If there were less than 2 patients per treatment arm within an investigative site, those data were pooled with other small investigative sites. All tests were performed at a two-tailed level of significance of 0.05, with no adjustments made for multiple comparisons.

3. Results

3.1. Patient baseline characteristics and disposition

After the study was completed it was discovered that 17.7% of the olanzapine group, 14.1% of the risperidone group, and 18.6% of the haloperidol group were not on antipsychotic medications prior to randomization. Additionally, approximately 25.8% of all patients were randomized to the same antipsychotic medication that they were prescribed prior to enrollment. Among patients who completed the study, the mean modal dose was 12.3 mg/day for olanzapine, 5.2 mg/day for risperidone, and 8.2 mg/day for haloperidol treatment groups. Significantly fewer patients treated with olanzapine (40.9%) or risperidone (48.1%)

compared to haloperidol (61.9%) used anticholinergic medications at any time during the trial (Fisher's exact test $p < 0.01$). The percentage of patients who reported ever using benzodiazepines during the trial was not significantly different between treatment groups (olanzapine = 37.1%; risperidone = 38.6% and haloperidol = 41.2%).

The percentages of patients who completed the study were as follows: 40.3% of olanzapine-treated patients, 34.2% of risperidone-treated patients and 27.8% of haloperidol-treated patients. Pairwise comparisons of the Kaplan–Meier survival curves showed a significant difference in time to discontinuation for any reason between olanzapine and haloperidol treatment groups (log-rank test $p = 0.02$) with no other comparisons reaching significance. The median (95% confidence interval) time to discontinuation for any reason was 220 (137, 254) days for olanzapine-treated patients, 145 (111, 225) days for risperidone-treated patients, and 94 (60, 169) days for haloperidol-treated patients. Reasons for discontinuation are presented in Table 2. Across all treatment groups, 61% of patients were considered to be compliant with their prescribed treatment during their participation in the study. Patients who missed 5 days or more of study medication or patients who took more than the prescribed dose on a regular basis were discontinued from the active comparison phase of the study.

Baseline characteristics were compared between patients randomized before and those randomized after dropping the haloperidol treatment arm. For patients randomized prior to dropping the haloperidol arm, 32% were female, 58% were Caucasian, mean age was 39.4 years old, and the mean PANSS total score was 82.1 at baseline. For patients randomized after dropping the haloperidol arm, 24% were female, 62% were Caucasian, mean age was 38.7 years old, and the mean PANSS Total score was 84.9 at baseline. These baseline characteristics were not significantly different when comparing patients randomized before versus after the haloperidol arm was dropped. However, mean PANSS positive score for patients randomized prior to dropping the haloperidol arm was significantly lower when compared to patients randomized after the haloperidol arm was dropped (20.8 vs 22.1, respectively; $t(412) = -2.7$, $p = 0.007$).

3.2. Neurocognitive efficacy

3.2.1. Sample composite LOCF

Based on the LOCF mean change from baseline for the neurocognitive composite based on the sample at baseline, there were no significant differences between any of the treatment groups at weeks 8, 24, or 52 (Table 3; Fig. 1). At all time points, there was significant improvement within all groups. At the 52-week endpoint, the z-scores based on the sample composite mean ± SD were 0.17 ± 0.51 for the olanzapine-treated group ($p < 0.01$), 0.18 ± 0.46 for risperidone-treated group ($p < 0.01$), and 0.12 ± 0.49 for haloperidol-treated group ($p = 0.04$).

3.2.2. Sample composite OC

Based on OC analysis at 52 weeks, change in the composite scores from the sample at baseline was significantly different between the haloperidol treatment group and both the olanzapine ($F(1,109) = 8.28$, $p < 0.01$) and risperidone ($F(1,109) = 4.28$, $p = 0.04$) treatment groups. There were no statistically significant differences between olanzapine- and risperidone-treated patients ($F(1,109) = 1.18$, $p = 0.28$). Mean change was significantly improved at endpoint within the olanzapine ($p < 0.01$) and risperidone ($p < 0.01$) treatment groups, while the haloperidol treatment group did not significantly improve ($p = 0.91$). The absence of a significant change in the haloperidol

Table 2
Patient disposition

Reason for discontinuation	Olz $N=159$ n(%)	Ris $N=158$ n(%)	Hal $N=97$ n(%)	p-values[a]			
				Olz vs. Ris	Olz vs. Hal	Ris vs. Hal	Overall
Adverse event	15 (9.4)	24 (15.2)	14 (14.4)	0.13	0.23	1.00	0.26
Satisfactory response	0	1 (0.6)	1 (1.0)	0.50	0.38	1.00	0.53
Lack of efficacy	20 (12.6)	18 (11.4)	16 (16.5)	0.86	0.46	0.26	0.50
Lost to follow-up	14 (8.8)	12 (7.6)	7 (7.2)	0.84	0.82	1.00	0.92
Patient decision	20 (12.6)	20 (12.7)	14 (14.4)	1.00	0.71	0.71	0.91
Physician decision	3 (1.9)	2 (1.3)	2 (2.1)	1.00	1.00	0.64	0.90
Criteria not met/Compliance	22 (13.8)	25 (15.8)	16 (16.5)	0.64	0.60	1.00	0.80
Sponsor decision	1 (0.6)	2 (1.3)	0	0.62	1.00	0.53	0.62
Protocol complete	64 (40.3)	54 (34.2)	27 (27.8)	0.30	0.06	0.33	0.13

Olz=olanzapine, Ris=risperidone, Hal=haloperidol.
[a] Frequencies analyzed using two-tailed Fisher's exact test.

Table 3
Neurocognitive composite score based on the sample's baseline data

Week	Therapy	*N*	Baseline mean ± SD	Endpoint mean ± SD	Within group *p*-value[a]	Change to endpoint		
						Mean ± SD	Pairwise *p*-value[b]	
8	Olz	130	0.04 ± 0.90	0.16 ± 0.93	<0.01	0.12 ± 0.42	O vs. R	0.81
	Ris	131	−0.02 ± 0.82	0.09 ± 0.82	<0.01	0.11 ± 0.44	O vs. H	0.72
	Hal	76	−0.01 ± 0.78	0.13 ± 0.78	<0.01	0.14 ± 0.41	R vs. H	0.58
24	Olz	132	0.03 ± 0.89	0.19 ± 0.96	<0.01	0.16 ± 0.49	O vs. R	0.99
	Ris	131	−0.02 ± 0.82	0.14 ± 0.83	<0.01	0.15 ± 0.48	O vs. H	0.73
	Hal	76	−0.01 ± 0.78	0.13 ± 0.81	0.01	0.14 ± 0.46	R vs. H	0.74
52	Olz	132	0.03 ± 0.89	0.20 ± 0.97	<0.01	0.17 ± 0.51	O vs. R	0.72
	Ris	131	−0.02 ± 0.82	0.17 ± 0.81	<0.01	0.18 ± 0.46	O vs. H	0.40
	Hal	76	−0.01 ± 0.78	0.11 ± 0.86	0.04	0.12 ± 0.49	R vs. H	0.25

Last-observation-carried-forward visitwise mean change to endpoint.
n=Number of patients having both baseline and endpoint.
Abbreviations: Olz=olanzapine, Ris=risperidone, Hal=haloperidol.
[a] Within group *p*-value based on t-test on mean change.
[b] Between group *p*-values are from Type III sum of squares ANOVA including age, baseline, therapy, and investigator.

group may have been partially attributable to one outlier, who worsened substantially from week 24 to week 52. Even when removing this outlier, the haloperidol treatment group did not significantly improve ($p=0.29$; mean change with outlier 0.01 vs without outlier 0.10).

Characteristics of patients who were randomized before the haloperidol arm was dropped and completed 52 weeks of treatment were compared to patients enrolled after the protocol changed when the study was completed. For these patients, the visitwise 52-week OC subanalysis conducted on the sample composite yielded similar results as above.

3.2.3. Normative composite LOCF

Based on LOCF mean change from baseline analysis at 52 weeks, the change in composite scores from the normative data was not significantly different between groups. The olanzapine and risperidone treatment groups significantly improved (both p's<0.01), with a trend level of change for the haloperidol treatment group ($p=0.08$).

3.2.4. Normative composite OC

Based on OC mean change from baseline analysis to 52 weeks, the change in composite scores from the normative data was significantly different between haloperidol-treated patients and both olanzapine- ($F(1,107)=4.66$, $p=0.03$) and risperidone- ($F(1,107)=5.18$, $p=0.03$) treated patients with no significant differences between olanzapine and risperidone treatment groups. Again there was significant within-group improvement for the olanzapine ($p<0.01$) and risperidone ($p<0.01$) treatment groups but not for haloperidol treatment group ($p=0.87$). Again, the absence of change ($p=0.28$) in the haloperidol group may have been partially attributable to the one outlier (mean change 0.04 vs 0.18 after the outlier was removed).

3.3. Individual neurocognitive domains

The 52-week LOCF mean change from baseline analyses for individual neurocognitive domains based on the sample's baseline data suggest that olanzapine-treated patients improved significantly more than haloperidol-treated patients in the processing speed ($F(1,307)=7.90$, $p<0.01$) and motor function domains ($F(1,306)=4.29$, $p=0.04$) (Fig. 2). There were no significant differences between risperidone and either olanzapine or haloperidol treatment groups on any of the individual neurocognitive domains. Olanzapine-treated patients significantly improved on all neurocognitive domains (all $p=0.04$) except visuospatial ability and verbal fluency. Risperidone-treated patients significantly improved on all domains (all $p<0.05$) except verbal fluency. Haloperidol-treated patients improved only on the learning and memory domain ($p<0.01$). A similar profile was found for the 52-week LOCF mean change from baseline analyses for neurocognitive domains based on normative data.

3.4. Clinical symptom efficacy

The ANOVA for baseline to week 8 LOCF mean change in PANSS scores (positive, negative, and total) revealed no statistically significant differences between treatment groups. The LOCF mean change to 52 weeks showed no statistically significant differences between olanzapine and risperidone treatment groups or risperidone and haloperidol treatment groups. However, at the 52-week endpoint, the olanzapine-treated patients had significantly more improve-

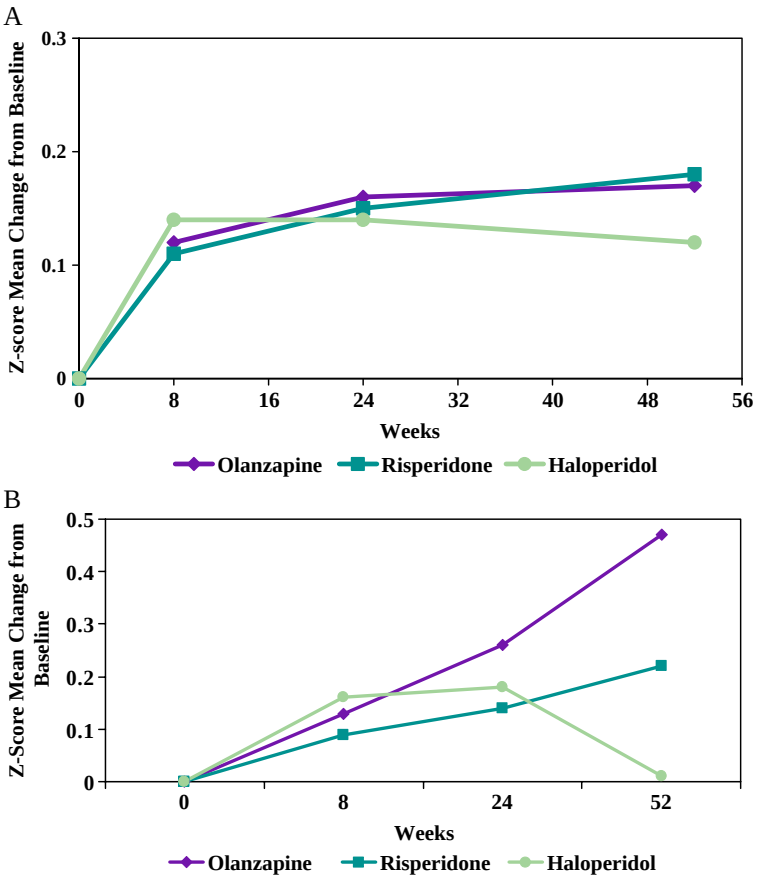


Fig. 1. Panel A. Visitwise Last-observation-carried-forward (baseline to 52-week endpoint): Z-score mean change composite scores based on the sample's baseline data in all randomized patients. Within group p-values at 52 weeks: olanzapine $p<0.001$; risperidone $p<0.001$; haloperidol $p=0.037$. Pairwise group p-values at 52 weeks: olanzapine vs risperidone $p=0.716$; olanzapine vs haloperidol $p=0.403$; risperidone vs haloperidol $p=0.251$. Panel B. Visitwise observed case (baseline to 52-week endpoint): Z-score mean change composite scores based on the sample's baseline data in all randomized patients. Within group p-values at 52 weeks: olanzapine $p<0.001$; risperidone $p<0.001$; haloperidol $p=0.907$. Pairwise group p-values at 52 weeks: olanzapine vs risperidone $p=0.280$; olanzapine vs haloperidol $p=0.005$; risperidone vs haloperidol $p=0.041$. Models include investigator, therapy, baseline, and age.

ment compared to haloperidol-treated patients on the PANSS positive items ($F(1355)=5.80$, $p=0.02$) and PANSS total scores ($F(1355)=9.04$, $p<0.01$) with a trend level significance for the PANSS negative items ($F(1355)=3.55$, $p=0.06$). All treatment groups significantly improved at 52 weeks LOCF on all three PANSS measurements ($p<0.02$). There were no statistical differences between any treatment groups for either the MADRS or HAMA (Table 4).

Among patients who responded, there were no significant group differences in relapse rates (olanzapine=13.3% (8/60); risperidone=26.6% (17/75); haloperidol=25.7% (9/35); $p=0.15$). For patients who stabilized, there was a significant group difference for relapse rates between olanzapine (11.6%; 15/129) and risperidone (22.3%; 27/121; $p=0.03$), with no significant differences with the haloperidol treatment group (22.2%; 16/72).

Correlations between LOCF mean change in sample cognition composite score and changes in PANSS positive, PANSS negative, PANSS total, and Simpson–Angus total score within each therapy group revealed significant correlations between the change in sample composite score and change in PANSS negative ($r=0.254$, $p=0.003$), PANSS total ($r=0.281$, p=0.001), and Simpson–Angus ($r=0.277$, $p=0.001$) in olanzapine-treated patients. There was a significant correlation between LOCF mean change in the sample cognition composite score and change in PANSS total ($r=0.183$, $p=0.037$) and change in Simpson–Angus total

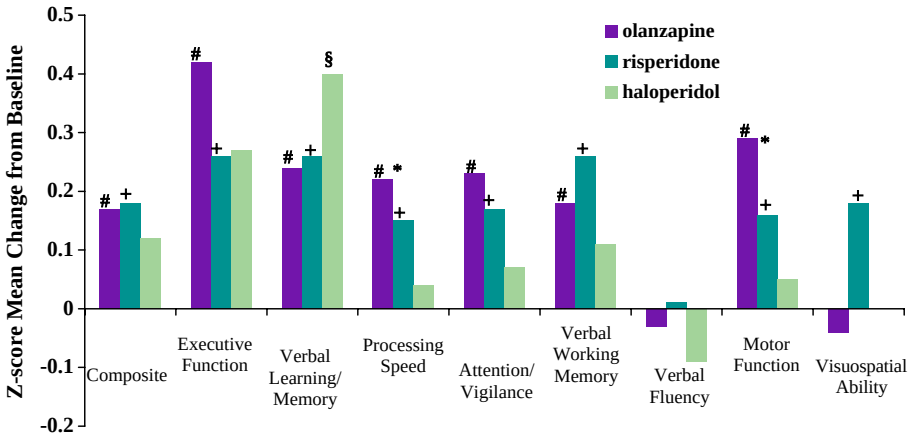


Fig. 2. Last-observation-carried-forward (baseline to 52-week endpoint): Z-score mean change neurocognitive domain scores based on the sample's baseline data in all randomized patients. * Significant difference between olanzapine and haloperidol treatment groups; # olanzapine treatment group improved significantly; + risperidone treatment group improved significantly; § haloperidol treatment group improved significantly.

score ($r=0.226$, $p=0.010$) in risperidone-treated patients. There were no significant correlations between these measures within the haloperidol-treated patients. When observed case mean changes were used, only the olanzapine treatment group had a significant correlation between mean change in sample cognition composite score and mean change in PANSS negative score ($r=0.328$, $p=0.011$).

3.5. Safety measures at 52-week endpoint

Adverse events that were reported in greater than 10% of any group or that were significantly different between any two groups are presented in Table 5. Weight, cholesterol, non-fasting glucose, prolactin, and triglycerides are presented in Table 6. Significantly more patients in the olanza-

Table 4
Clinical symptom assessments mean ± standard deviation (SD) baseline and last-observation-carried-forward visitwise mean change from baseline to week 52

	Therapy	Baseline Mean ± SD	Change to Endpoint					
			n	Mean ± SD	*p*-values[a]			
					Olz vs. Ris	Olz vs. Hal	Ris vs. Hal	Overall
PANSS total score	Olz	82.6 ± 13.1	153	−12.4 ± 16.0	0.11	<0.01	0.12	0.01
	Ris	84.1 ± 14.7	148	−9.5 ± 15.5				
	Hal	82.7 ± 14.1	94	−7.6 ± 16.3				
PANSS positive score	Olz	21.2 ± 4.4	153	−4.3 ± 4.9	0.18	0.02	0.22	0.15
	Ris	21.4 ± 4.9	148	−3.6 ± 5.5				
	Hal	21.4 ± 4.5	94	−3.1 ± 5.8				
PANSS negative score	Olz	20.6 ± 5.3	153	−2.5 ± 5.3	0.16	0.06	0.52	0.04
	Ris	21.0 ± 5.3	148	−1.6 ± 4.9				
	Hal	20.7 ± 5.5	94	−1.5 ± 4.8				
MADRS	Olz	13.2 ± 8.7	153	−2.9 ± 8.3	0.42	0.09	0.33	0.45
	Ris	14.1 ± 9.3	150	−2.2 ± 8.7				
	Hal	14.4 ± 10.2	92	−1.7 ± 7.9				
HAMA	Olz	9.6 ± 6.3	153	−2.1 ± 5.3	0.49	0.35	0.74	0.37
	Ris	10.2 ± 6.6	150	−1.6 ± 6.1				
	Hal	10.9 ± 7.0	92	−1.5 ± 6.1				

Abbreviations: Olz=olanzapine, Ris=risperidone, Hal=haloperidol.
[a] Means analyzed using Type III sum of squares from analysis of variance including treatment, investigator, and treatment-by-investigator interaction.

pine (13.8%) and risperidone (10.8%) treatment groups relative to the haloperidol treatment group (2.1%; $p < 0.01$ and $p = 0.01$, respectively) reported weight gain as an adverse event. The olanzapine treatment group had a significantly greater increase in weight than the risperidone ($F(1,356) = 7.65$, $p < 0.01$) and haloperidol ($F(1,356) = 43.41$, $p < 0.01$) treatment groups. Risperidone-treated patients gained significantly more weight than haloperidol-treated patients ($F(1,356) = 17.21$, $p < 0.01$). Triglycerides mean change was significantly higher for olanzapine treatment group relative to risperidone treatment group ($F(1,196) = 6.42$, $p = 0.01$) with no other significant differences between olanzapine- and haloperidol-treated patients or risperidone- and haloperidol-treated patients. Cholesterol mean change was significantly higher for the olanzapine treatment group than both risperidone ($F(1,257) = 8.90$, $p = 0.03$) and haloperidol ($F(1,257) = 9.23$, $p = 0.03$) treatment groups. Prolactin mean change was significantly higher for the risperidone versus olanzapine treatment group ($F(1,309) = 69.82$, $p < 0.01$) and for the risperidone versus haloperidol treatment group ($F(1,309) = 28.39$; $p < 0.01$), while there was no statistical difference between the olanzapine and haloperidol treatment groups ($p = 0.07$). Non-fasting glucose mean change was not significantly different between any treatment groups.

On the AIMS total, haloperidol-treated patients had significantly greater mean change than both olanzapine- ($F(1,355) = 5.27$, $p = 0.02$) and risperidone-treated patients ($F(1,355) = 4.96$, $p = 0.03$) with no significant difference between olanzapine and risperidone treatment groups ($p = 0.95$; Table 6). Similarly, for the Barnes Global mean change score, haloperidol-treated patients significantly worsened compared to both the olanzapine- ($F(1,356) = 14.05$, $p < 0.01$) and risperidone- ($F(1,356) = 7.45$, $p < 0.01$) treated groups with no significant difference between olanzapine and risperidone treatment patients ($p = 0.25$). In parallel, significantly fewer patients in the olanzapine- (8.8%) and risperidone-treated groups (12.7%) relative to the haloperidol-treated group (24.7%; $p < 0.01$ and $p < 0.02$, respectively) reported akathisia as an adverse event. There were no significant differences for the Simpson–Angus total mean change scores between any of the treatment groups.

4. Discussion

For the protocol-specified primary mean change to baseline to LOCF endpoint analysis, all three pharmacotherapies, olanzapine, risperidone, and haloperidol,

Table 5
Treatment-emergent adverse events in >10% of any group or significantly different between groups

	Olz *n*(%)	Ris *n*(%)	Hal *n*(%)	*p*-values[a]			
				Olz vs. Ris	Olz vs. Hal	Ris vs. Hal	Overall
Somnolence	41 (25.8)	31 (19.6)	18 (18.6)	0.23	0.22	0.87	0.30
Depression	28 (17.6)	31 (19.6)	11 (11.3)	0.67	0.21	0.12	0.21
Headache	27 (17.0)	40 (25.3)	20 (20.6)	0.08	0.51	0.45	0.20
Insomnia	26 (16.4)	36 (22.8)	20 (20.6)	0.16	0.41	0.76	0.34
Anxiety	25 (15.7)	28 (17.7)	16 (16.5)	0.65	0.86	0.87	0.88
Nausea	24 (15.1)	19 (12.0)	13 (13.4)	0.51	0.86	0.85	0.73
Weight gain	22 (13.8)	17 (10.8)	2 (2.1)	0.50	<0.01	0.01	<0.01
Pain	20 (12.6)	24 (15.2)	7 (7.2)	0.52	0.21	0.08	0.16
Rhinitis	19 (11.9)	31 (19.6)	19 (19.6)	0.07	0.11	1.00	0.12
Hallucinations	18 (11.3)	18 (11.4)	11 (11.3)	1.00	1.00	1.00	1.00
Nervousness	18 (11.3)	23 (14.6)	17 (17.5)	0.41	0.19	0.60	0.36
Dry mouth	17 (10.7)	14 (8.9)	11 (11.3)	0.71	1.00	0.52	0.80
Diarrhea	15 (9.4)	18 (11.4)	15 (15.5)	0.59	0.16	0.35	0.34
Dizziness	15 (9.4)	16 (10.1)	16 (16.5)	0.85	0.11	0.17	0.21
Akathisia	14 (8.8)	20 (12.7)	24 (24.7)	0.28	<0.01	0.02	<0.01
Tremor	13 (8.2)	11 (7.0)	13 (13.4)	0.83	0.20	0.12	0.21
Paranoid reaction	10 (6.3)	14 (8.9)	10 (10.3)	0.41	0.34	0.83	0.48
Abnormal thinking	10 (6.3)	16 (10.1)	6 (6.2)	0.23	1.00	0.36	0.40
Vomiting	10 (6.3)	18 (11.4)	13 (13.4)	0.12	0.07	0.69	0.12
Agitation	7 (4.4)	17 (10.8)	8 (8.2)	0.04	0.27	0.67	0.09
Constipation	6 (3.8)	18 (11.4)	6 (6.2)	0.01	0.38	0.19	0.03

[a] Frequencies analyzed using Fisher's exact test by including only those patients with at least one postbaseline assessment including treatment, investigator, and treatment-by-investigator interaction.

Table 6
Safety and laboratory mean ± standard deviation (SD) baseline and mean change from baseline to 52-week endpoint

Variables	Therapy	*n*	Baseline Mean ± SD	Change to endpoint Mean ± SD	*p*-values[a]			
					Olz vs. Ris	Olz vs. Hal	Ris vs. Hal	Overall
Weight (kg)	Olz	152	88.14 ± 20.38	3.00 ± 7.87				
	Ris	150	86.14 ± 18.65	0.81 ± 6.26	<0.01	<0.01	<0.01	<0.01
	Hal	94	88.82 ± 17.94	−2.34 ± 4.80				
Cholesterol (mg/dL)	Olz	111	175.35 ± 71.01	8.97 ± 31.41				
	Ris	113	171.82 ± 83.71	−2.38 ± 25.38	<0.01	<0.01	0.66	0.11
	Hal	71	171.32 ± 79.59	−3.72 ± 24.41				
Glucose, non-fasting (mg/dL)	Olz	111	90.42 ± 46.90	10.44 ± 51.46				
	Ris	113	93.88 ± 56.34	1.19 ± 34.27	0.08	0.16	0.92	0.67
	Hal	71	93.18 ± 51.92	0.60 ± 57.26				
Prolactin (ng/mL)	Olz	136	23.92 ± 24.95	−9.73 ± 23.15				
	Ris	130	22.87 ± 27.41	18.75 ± 31.23	<0.01	0.07	<0.01	<0.01
	Hal	81	20.18 ± 24.85	−2.37 ± 25.84				
Triglycerides (mg/dL)	Olz	96	160.51 ± 117.02	31.58 ± 131.61				
	Ris	95	216.82 ± 280.54	−34.04 ± 222.25	0.01	0.10	0.84	0.14
	Hal	40	156.15 ± 158.63	−22.27 ± 117.38				
AIMS Total Score	Olz	152	1.70 ± 2.95	0.09 ± 2.91				
	Ris	150	2.24 ± 3.89	0.12 ± 3.67	0.95	0.02	0.03	0.23
	Hal	93	2.69 ± 3.79	1.13 ± 4.36				
Barnes Global Score	Olz	153	0.67 ± 0.94	−0.18 ± 0.92				
	Ris	150	0.65 ± 1.01	0.01 ± 1.13	0.25	<0.01	<0.01	<0.01
	Hal	93	0.75 ± 0.99	0.41 ± 1.26				
Simpson–Angus Total Score	Olz	153	1.85 ± 2.91	−0.73 ± 2.92				
	Ris	149	1.81 ± 2.88	−0.06 ± 3.16	0.06	0.21	0.71	0.08
	Hal	94	2.28 ± 3.76	−0.22 ± 3.65				

Olz=Olanzapine, Ris=Risperidone, Hal=Haloperidol.
n=Total number of patients in each treatment group having both baseline and postbaseline measurements.
[a] Means analyzed using Type III sum of squares from analysis of variance including treatment, investigator, and treatment-by-investigator interaction.

were equivalent with regards to neurocognitive improvement at 8, 24, and 52 weeks of treatment. Neurocognitive composite scores based on either the baseline sample or on normative data improved significantly at all time points in the olanzapine and risperidone treatment groups but not the haloperidol treatment group. For the protocol-specified secondary analysis, the olanzapine-treated patients improved significantly more than the haloperidol-treated patients, but not the risperidone-treated patients on the PANSS positive and total at the LOCF 52-week endpoint but not the 8-week endpoint. There were no significant differences in PANSS improvement between the risperidone- and haloperidol-treated patients or between olanzapine- and risperidone-treated patients.

These results suggest that the difference between atypical medications and typicals on cognitive deficits might not be as large as initially reported (Purdon et al., 2000; Keefe et al., 1999). In contrast to many previous studies (reviewed by Harvey and Keefe, 2001; Woodward et al., 2005), this study had several methodological advances. This study used fair comparisons with dosages aligned with current recommendations, non-treatment-resistant patients, and an extensive cognitive battery. The sample size of 414 was large, it included a double-blind randomized design not found in many of the studies completed to date, and the study duration of one-year was substantial.

Perhaps the greatest methodological advantage of this study over many previous studies was the medication dosing. Patients randomized to the haloperidol arm received a mean modal dose of 8.2 mg per day (410 mg chlorpromazine-equivalent). This dose is substantially lower than the mean modal dose of typical antipsychotic medication (736 mg chlorpromazine-equivalent) used as a comparator with atypical antipsychotics in previous studies investigating

neurocognitive improvement (Harvey and Keefe, 2001). Although not statistically significant from zero, the neurocognitive improvement in patients receiving relatively low doses of haloperidol in this study suggests that the neurocognitive advantage of atypical antipsychotics is partially attributable to the high doses of typical medication usually prescribed in these studies. Although methodological differences between this study and the studies of low-dose haloperidol (Green et al., 2002; Keefe et al., 2004; Harvey, abstract from ICSR, 2003) limit direct comparison, the effect sizes of neurocognitive change with haloperidol were approximately 0.2 in each of the four studies. The convergence of these findings suggests that when administered at relatively low doses, haloperidol may yield a small neurocognitive benefit.

Three limitations of this study warrant discussion. First, at baseline, about 18% of olanzapine-treated patients and 14% of risperidone-treated patients were being treated with olanzapine or risperidone, respectively, and overall 26% were randomized to their previous therapy. This design flaw may have reduced the ability of the study to detect the full impact of treatment. Thus, the neurocognitive and symptom benefit reported here may underestimate the actual benefit of beginning a new treatment in clinical practice, in which patients are almost always given medications on which they had not recently failed. One method of reducing the impact of previous treatment would have been to establish a drug-free washout period (Purdon et al., 2000), but this strategy was not taken due to ethical concerns of withholding medication. Second, the option to drop the haloperidol arm of the study due to recruitment difficulties was described in the formal protocol. However, this decision created a different circumstance under which patients were entered into the trial. It is possible that the variability created by this change reduced the sensitivity of the study to detect clinical differences. Third, the significant differences between haloperidol treatment and the two atypical antipsychotic treatments reported in the observed cases analyses depended upon an unexpected average neurocognitive worsening of the haloperidol-treated patients at the end of the study. The causes of this worsening have not been determined. As this is one of the few studies to report a potential neurocognitive benefit of haloperidol treatment, it may be that the neurocognitive benefit of haloperidol treatment is temporary or limited to patients who dropped out of the study for other reasons. As noted in the results, a substantial portion of the worsening in the haloperidol group was due to a single patient. However, previous work has suggested that patients who show neurocognitive benefit while receiving typical antipsychotics are limited to those who have greater symptom reduction and fewer side effects (Keefe et al., 2004), and would thus be less likely to leave the study.

For the individual neurocognitive domains, specifically domains of processing speed and motor function, the olanzapine treatment group improved to a greater extent than the haloperidol treatment group. None of the changes in individual neurocognitive domains were significantly different between olanzapine- and risperidone-treated patients. Olanzapine treatment was characterized by specific improvements in domains of executive function, learning and memory, processing speed, attention/vigilance, verbal working memory, and motor function. Risperidone treatment was characterized by all of these in addition to the domain of visuospatial memory. For haloperidol-treated patients, only the domain of memory and learning was significantly improved at 52 weeks. These patterns of improvement suggest that atypical antipsychotic treatment may particularly benefit tests measuring the generation of action and attention, and are consistent with previous reports of specific benefit for olanzapine treatment on attention and processing speed in first episode (Keefe et al., 2004) and treatment-resistant schizophrenia (Bilder et al., 2002).

The data from this study fail to replicate the differences reported by Purdon et al. (2000), who compared olanzapine to risperidone in fewer patients with a dose of risperidone that was high (6.1 mg) relative to current standards, especially in early phase patients. The results of the current study are more consistent with another large-sample study comparing olanzapine and risperidone (Harvey et al., 2003). The confluence of findings from these large-sample studies seems to highlight the similarities of risperidone and olanzapine in improving cognition, rather than their differences.

Significantly more patients treated with olanzapine or risperidone relative to haloperidol gained weight

and reported it as an adverse event. Relative to the risperidone treatment group, the olanzapine treatment group had significantly greater mean changes in weight, cholesterol and triglycerides. Relative to haloperidol treatment group, the olanzapine treatment group had significantly greater mean change in cholesterol. The olanzapine treatment group had a decrease in prolactin levels while the risperidone treatment group had an increase in prolactin levels. Relative to haloperidol treatment group, the risperidone treatment group had significantly greater mean change in levels of prolactin. All safety measurements were non-fasting, and thus could have been affected by recent food intake, which may have been especially important for triglyceride measurements (Uiterwaal et al., 1994). Both the AIMS and Barnes total scores significantly increased for the haloperidol treatment group relative to both olanzapine and risperidone treatment groups. The relative importance of the various side effects of antipsychotic medications on quality of life is controversial, and varies across patients (Ritsner et al., 2002) although atypicals appear in general to have a more favorable impact on subjective quality of life (Awas and Voruganti, 1999).

In conclusion, our hypothesis that olanzapine treatment would produce greater neurocognitive benefit than risperidone or haloperidol treatment for patients with schizophrenia was not confirmed. In fact, LOCF mean change analyses suggested that the three medications induced in this study were not significantly different in their neurocognitive efficacy. Protocol-specified secondary analyses suggested that olanzapine treatment was more effective than haloperidol treatment in reducing symptoms after up to 52 weeks of treatment, while risperidone treatment was not differentiated from olanzapine or haloperidol treatments. Additional post hoc analyses of the individual cognitive domains suggested that risperidone and olanzapine were more likely to provide significant improvement in specific areas of cognitive function.

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